

Phys 115B HW 2

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Question 1. *More Practice with Spherical Harmonics*

Consider a particle in three dimensions in a state

$$\psi(x, y, z) = A(x + y + 2z)e^{-\alpha r}$$

where $\alpha > 0$ is a positive constant, and A a normalization factor.

(a) Write the spherical harmonics $Y_1^{0,\pm 1}$ in terms of x, y, z , and

$$r = \sqrt{x^2 + y^2 + z^2}.$$

(b) Using your result from part (a), show that ψ can be written as a product of a purely r -dependent wave function and a linear combination of spherical harmonics.

Proof.

□

Question 2. Radial Momentum

In lecture, we defined an operator corresponding to the momentum in the radial direction,

$$p_r \equiv \frac{1}{2}(\hat{r} \cdot p + p \cdot \hat{r}) = -\frac{i\hbar}{2} \left(\frac{1}{r} r \cdot \nabla + \nabla \cdot \frac{r}{r} \right).$$

- (a) Show that $\hat{r} \cdot p$ is not Hermitian, and hence cannot itself be identified with momentum in the radial direction.
- (b) Using the definition of p_r above, show that

$$p_r^2 = -\hbar^2 \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right).$$

Proof.

□

Question 3. *The Size of Hydrogen (Griffiths 4.15, modified, 4.13 in second edition, still modified). Do parts (a)–(c) from the Griffiths problem.*

(a) Find $\langle r \rangle$ and $\langle r^2 \rangle$ for an electron in the ground state of hydrogen. Express your answers in terms of the Bohr radius.

(b) Find $\langle x \rangle$ and $\langle x^2 \rangle$ for an electron in the ground state of hydrogen.

(Hint: This requires no new integration - note that $r^2 = x^2 + y^2 + z^2$, and exploit the symmetry of the ground state).

(c) Find $\langle x^2 \rangle$ in the state $n = 2, \ell = 1, m = 1$.

(Hint: this state is not symmetrical in x, y, z . Use $x = r \sin(\theta) \cos(\phi)$).

(d) Find $\langle r \rangle$ and $\langle r^2 \rangle$ for an electron in a circular orbit of hydrogen with arbitrary principal quantum number n (corresponds to $\ell = n - 1$ and any allowed m).

(e) Compute the RMS uncertainty

$$\sqrt{\langle r^2 \rangle - \langle r \rangle^2}$$

in terms of r for the electron in part (d). Note that the fractional spread in r decreases with increasing n (in this sense the system “begins to look classical” for large n). How much more volume does a hydrogen atom in the $n = 100$ state occupy compared to the hydrogen atom in the ground state. (Hint — you might want to look at Griffiths 4.55, or 4.15 in the second edition.)

Proof.

□

Question 4. *The Size of Hydrogen continued (Griffiths 4.16, 4.14 in the second edition)*

What is the most probable value of r , in the ground state of hydrogen? (The answer is not zero!)

Hint: First you must figure out the probability that the electron would be found between r and $r + dr$.

Proof.

□

Question 5. *Where is the Electron (Griffiths 4.54, 4.45 in the second edition)*

What is the probability that an electron in the ground state of hydrogen will be found inside the nucleus?

- (a) First calculate the exact answer, assuming the wave function is correct all the way down to $r = 0$. Let b be the radius of the nucleus.

Here is the wave function:

$$\psi_{1,0,0}(r, \theta, \phi) = \frac{1}{\sqrt{\pi a^3}} e^{-r/a}$$

- (b) Expand your result as a power series in the small number $\epsilon := \frac{2b}{a}$, and show that the lowest-order term is the cubic: $p \approx \frac{4}{3} \left(\frac{b}{a}\right)^3$. This should be a suitable approximation, provided that $b \ll a$ (which it is).
- (c) Alternatively, we might assume that $\psi(r)$ is essentially constant over the (tiny) volume of the nucleus, so that $P \approx \frac{4}{3}\pi b^3 |\psi(0)|^2$. Check that you get the same answer this way.
- (d) Use $b \approx 10^{-15}$ m and $a \approx 0.5 \cdot 10^{-10}$ m to get a numerical estimate for P . Roughly speaking, this represents the “fraction of its time that the electron spends inside the nucleus”.

Proof.

□